



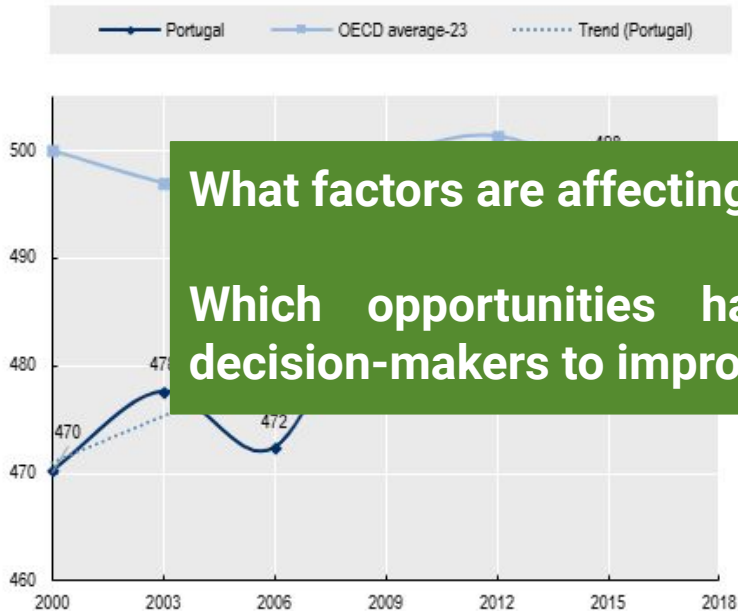
# Student performance prediction



Sabra Mabrouk



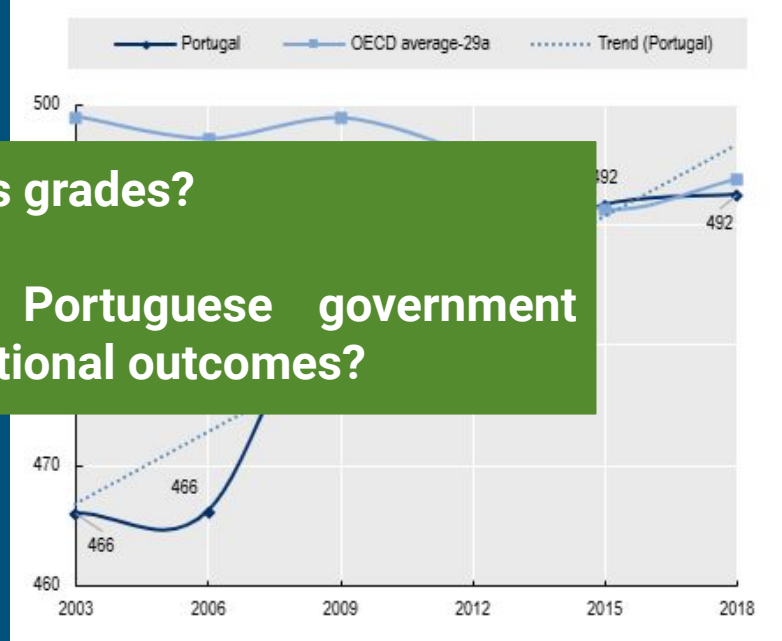
# Problem Statement



**What factors are affecting students grades?**

**Which opportunities have the Portuguese government decision-makers to improve educational outcomes?**

Source: OECD (2019[20]), PISA 2018 Results (Volume I): What Students Know and Can Do, Table I.B1.10, <https://doi.org/10.1787/5f07c754-en>



Source: OECD (2019[20]), PISA 2018 Results (Volume I): What Students Know and Can Do, Tables I.B1.11, <https://doi.org/10.1787/5f07c754-en>.

# Dataset

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- Population : 2 Portuguese secondary schools
- Period : 2005 - 2006
- Courses : Math & Portuguese

## Demographic and social data

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Age, sex, address, family size, parent's cohabitation status,, Internet access at home, with a romantic relationship, mother job, father job, student's guardian, mother's education, father's education, home to school travel time, quality of family relationships, free time after school, going out with friends, workday alcohol consumption, weekend alcohol consumption, current health status.

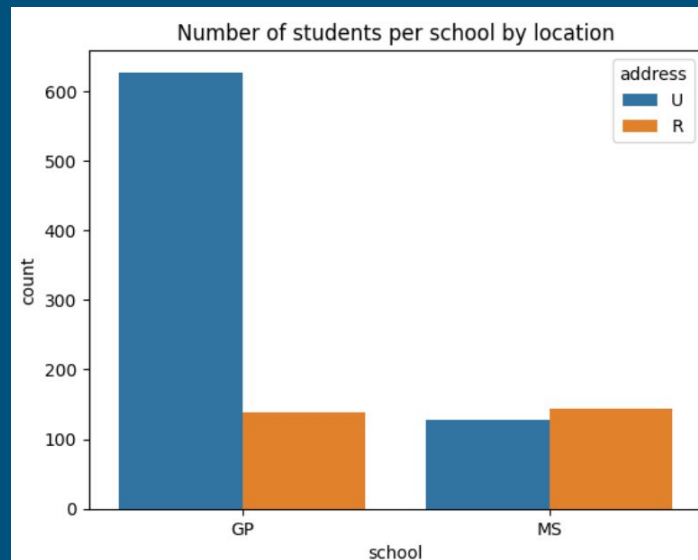
## School related data

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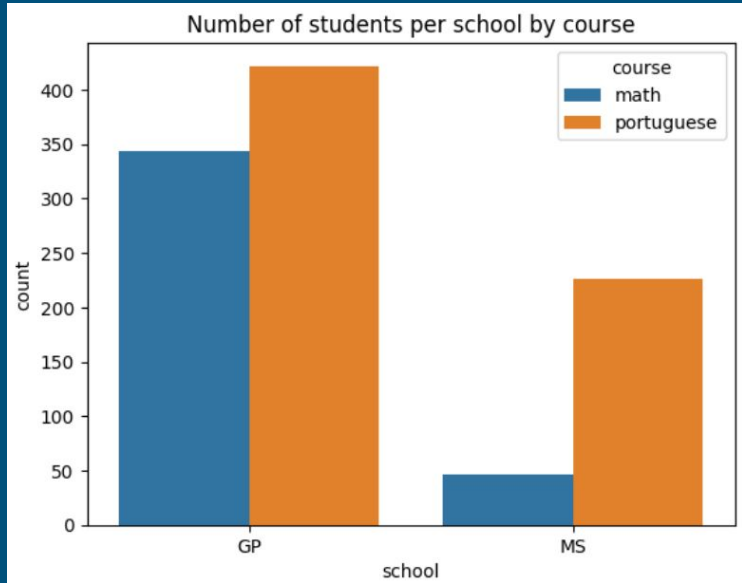
School, school support, family support, paid support, extra-curricular activities, attended nursery school, wants to take higher education, reason to choose this school, weekly study time, number of absences, number of failures, past grades.

# Exploratory Data Analysis

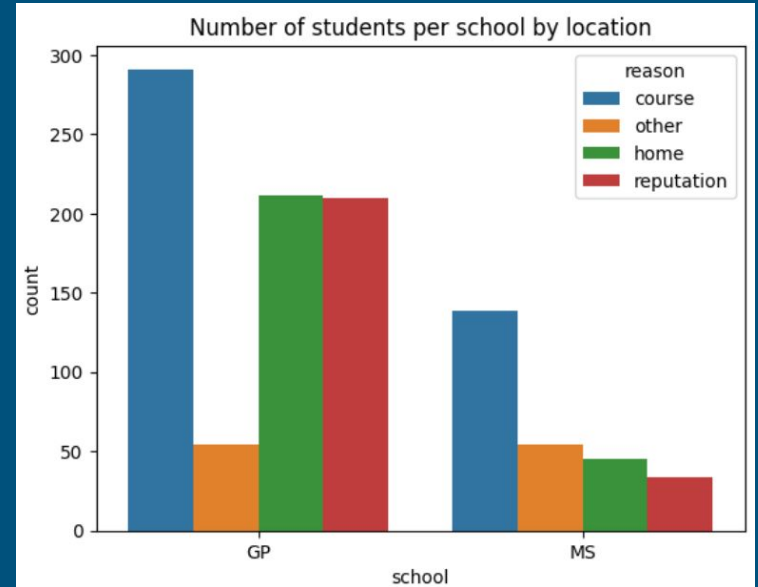
- Most of the students are from Gabriel Pereira (GP) School.
- Students at the GP School come mainly from urban areas.
- MS School enrolls a balanced proportion of students from urban and rural areas.



# Exploratory Data Analysis



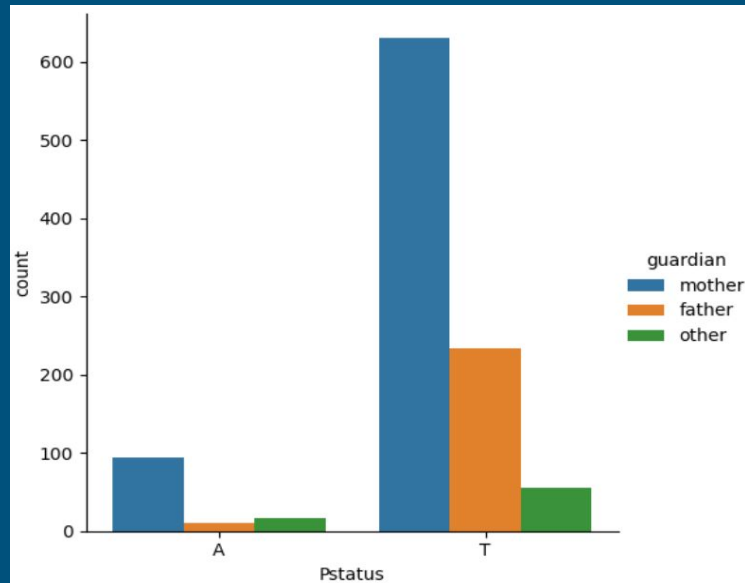
Most of the data relates to the Portuguese course at the two schools.



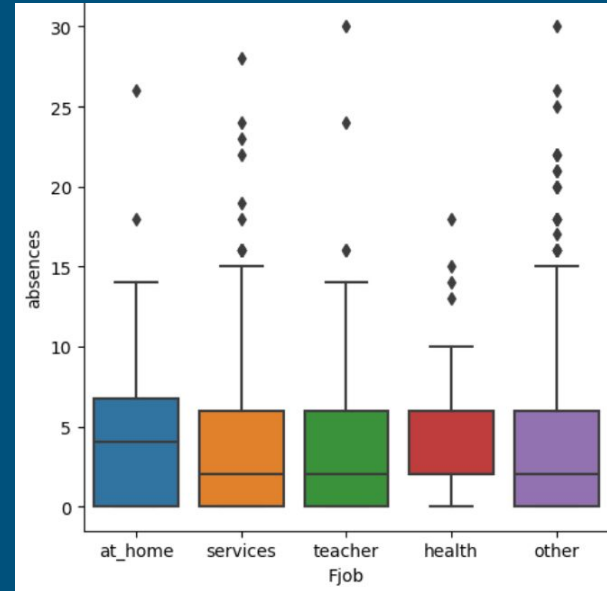
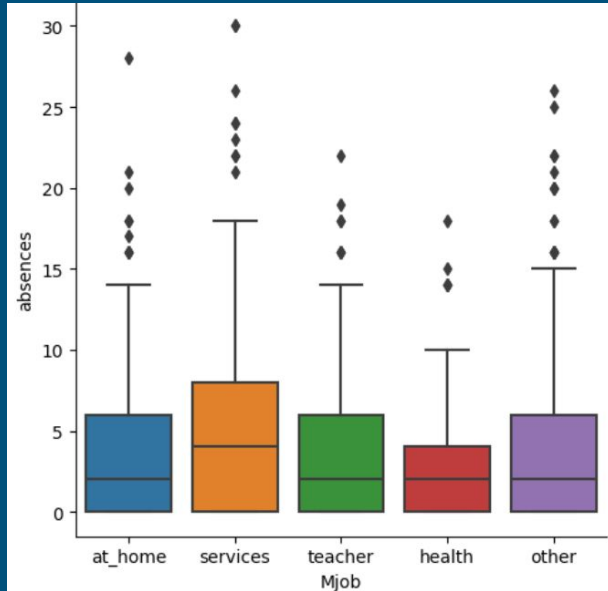
Students' choice of school was mainly based on the courses

# Exploratory Data Analysis

- Fewer students are from separated families in the data
- Most students are guarded by their mothers



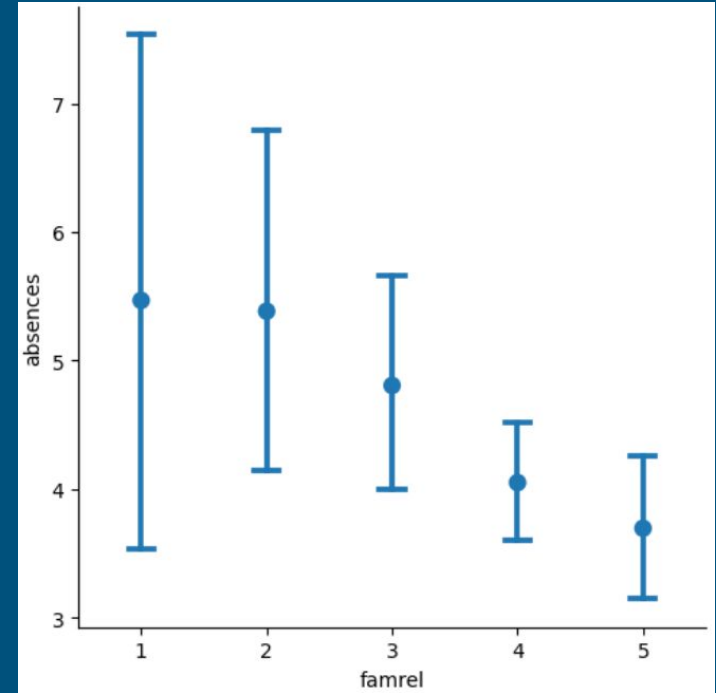
# Exploratory Data Analysis



The mother's job has a more effect on student absences than the father's

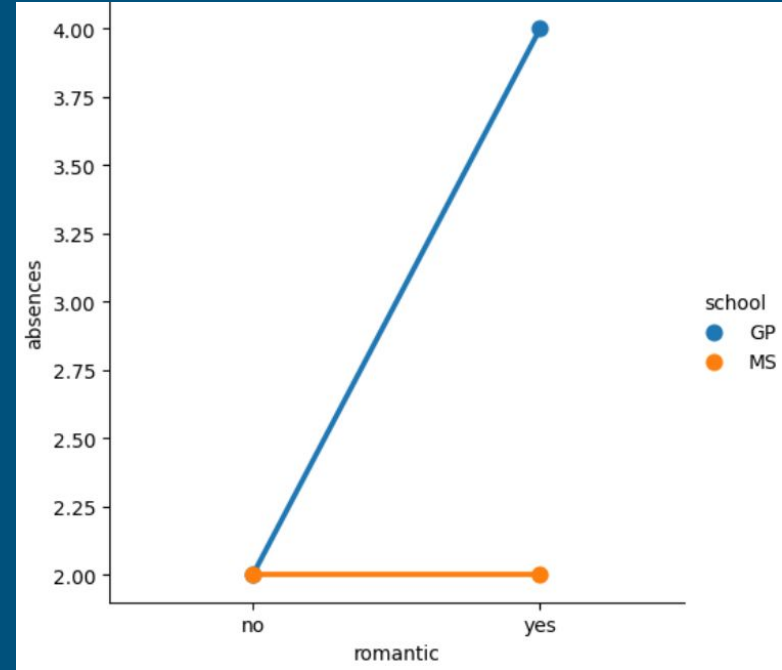
# Exploratory Data Analysis

- The average number of absences is slightly lower among students with better family relationships.
- The wide confidence intervals indicate that we cannot be sure that there is a true association



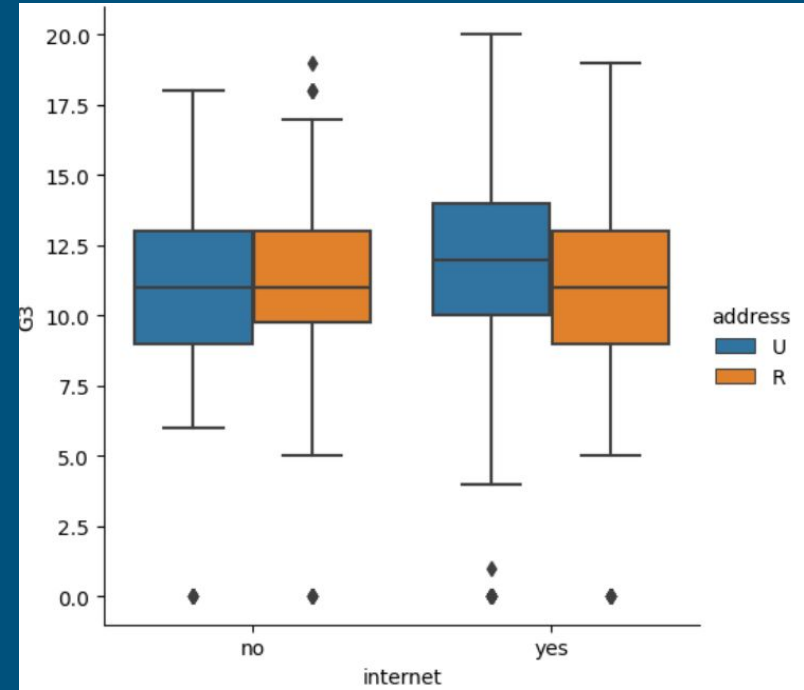
# Exploratory Data Analysis

- Students in a romantic relationship have a higher median number of absences in GP school, but this association does not hold true in MS school



# Exploratory Data Analysis

- Internet access brings no benefit to students from rural areas.
- A slight improvement is observed in the results of students from urban areas.



# Modeling

## Preprocessing

- Encoding categorical data
- Train/Test Splitting
  - 70% Training
  - 30% Test
- Scaling the data

## Models training

- Fitting 6 models
- Cross validation

## Evaluation

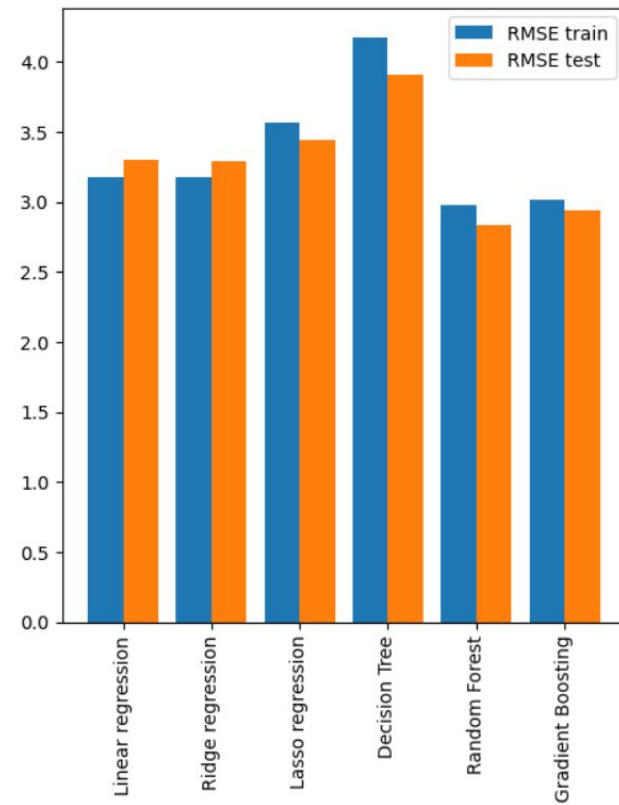
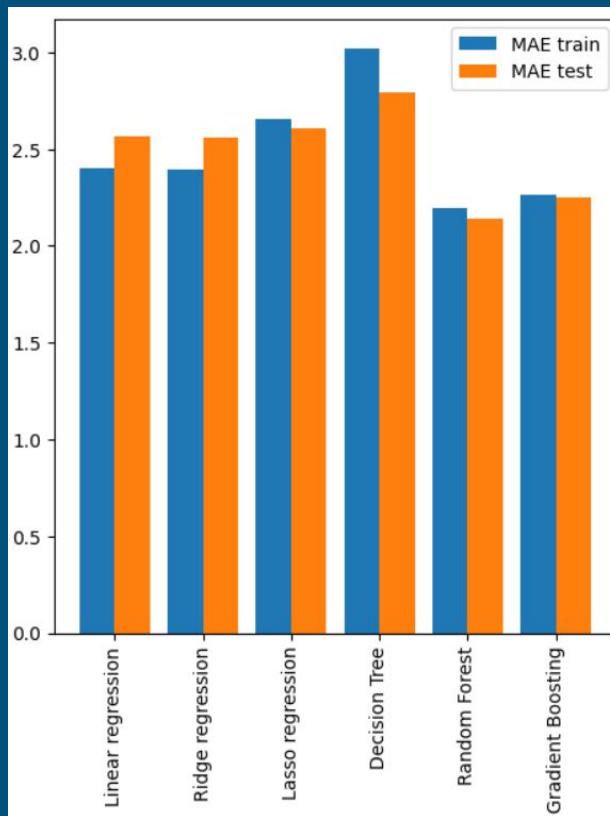
- 2 Metrics
  - Mean absolute error (MAE)
  - Root mean squared error (RMSE)

## Model Selection and Hyperparameters tuning

- Tuning the best 2 models
- Evaluation
- Features impact analysis

# Modeling

- Used models
  - Linear regression
  - Lasso
  - Ridge
  - Decision Tree
  - Random Forest
  - Gradient Boosting

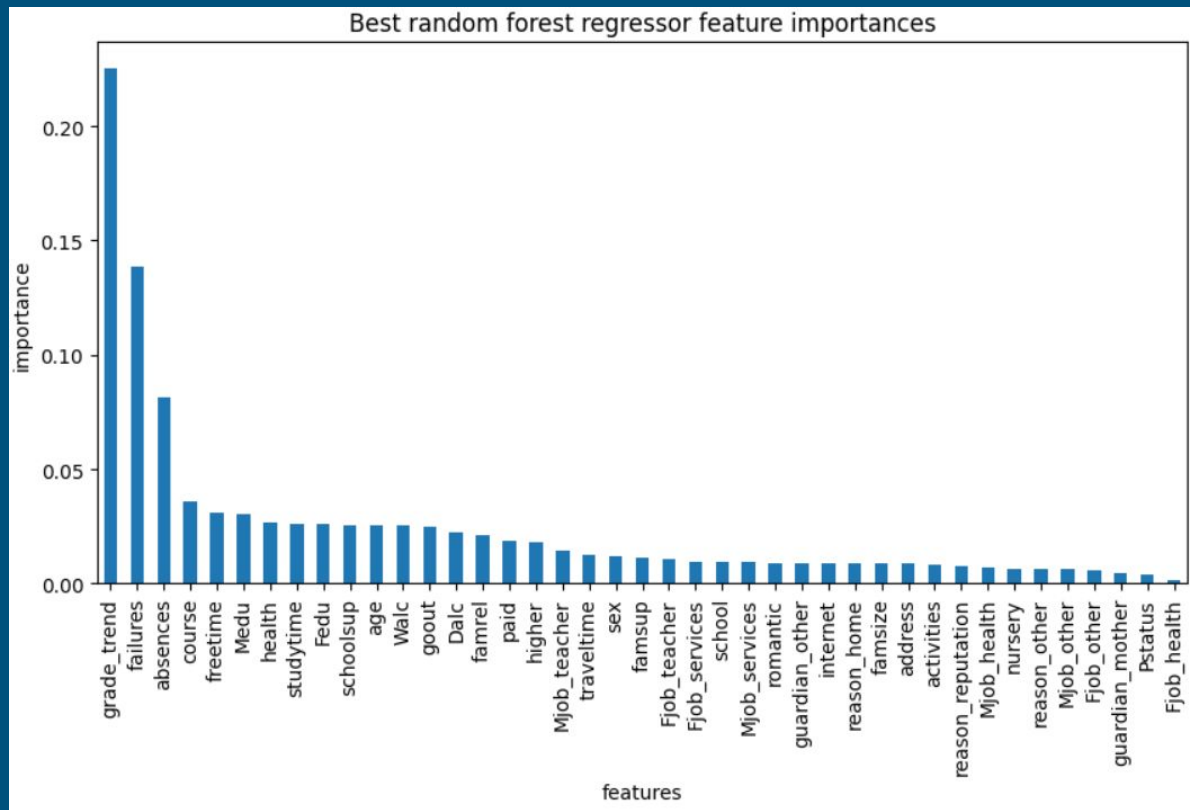


# Modeling

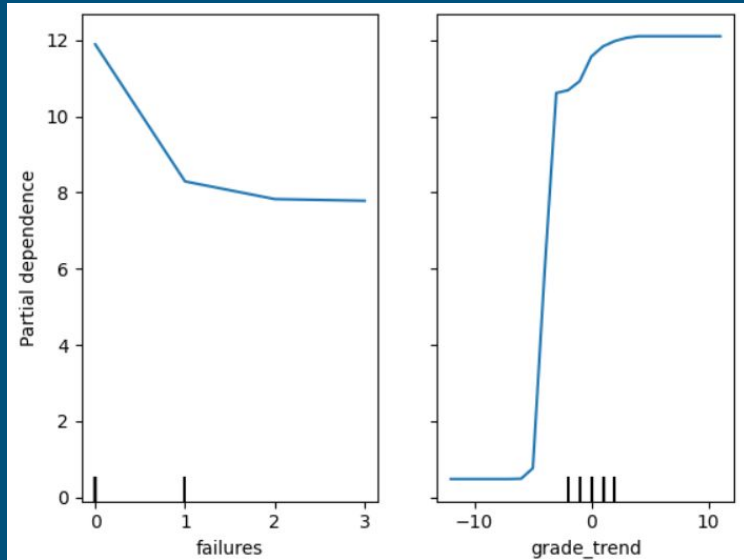
- Best 2 models
  - Random Forest: criterion=squared\_error, max\_depth=20, n\_estimators=100
  - Gradient Boosting: {'learning\_rate': 0.25, 'loss': 'squared\_error', 'max\_depth': 3, 'n\_estimators': 50}
  -

| Model             | Mean MAE train | Std MAE train mean | MAE test | Mean RMSE train | Std RMSE train | RMSE test |
|-------------------|----------------|--------------------|----------|-----------------|----------------|-----------|
| Random Forest     | 2.196709       | 0.182938           | 2.140584 | 2.978908        | 0.254237       | 2.833533  |
| Gradient Boosting | 2.252657       | 0.082890           | 2.264966 | 2.993485        | 0.119121       | 2.942300  |

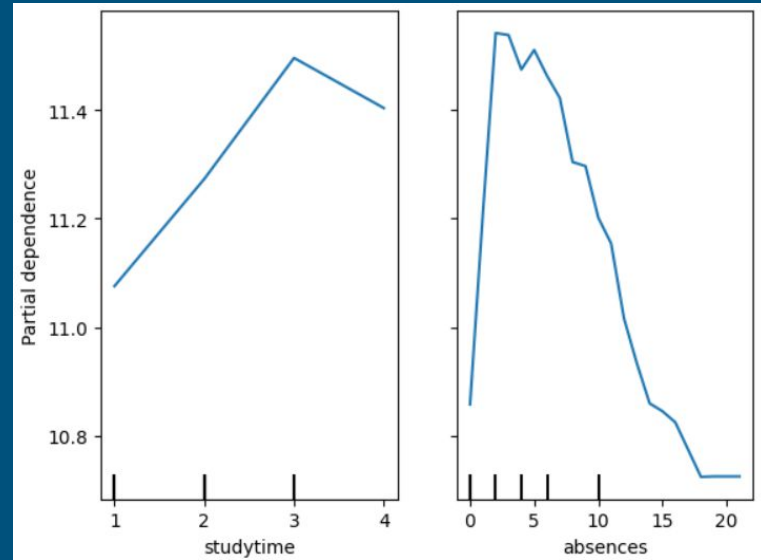
# Features impact



# Features impact



- Grade trend (G2-G1)
- Number of failures



- Weekly study time : 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, 4 - >10 hours
- Number of absences

# Scenarios analysis

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- Scenario 1

- Work on first and second year programs to ensure similar results.
- Establish a strategy to reduce the number of student failures by 80%

**Increase in success rate of about 11%**

- Scenario 2

- Imposing severely restrictive limits on the number of absences (No more than 3 per course).
- Encourage students to spend 5 to 10 hours per week on their studies (homework, projects,...).

**Increase in success rate of above 7%**

# Limitations

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- Limited dataset : Used only 2 schools and 2 years of data (2005- 2006)
- Potential problem of high dimensionality: 1038 observations vs 42 features
- Too much categorical variables making feature engineering very challenging
- Problem in the representation of combinations
  - Possible combinations of values for many categorical variables increase.
  - Underrepresented combinations may lead to biased predictions

# Conclusion

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- The most efficient of the 6 trained models in terms of MAE and RMSE was the random forest.
- Random Forest : prediction of the final grade with a MAE of 2.14 test set.
- Future work: Automatic feature selection methods can be explored to remove irrelevant features.

Thank You!